

Day 16 Review Sheet · Days 13-15

REVIEW

MathCraft · Grade 6 Mathematics · Professor Xavier Honablue, M.Ed.

Name: _____ Date: _____ Hour: _____

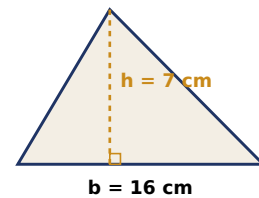
Covers: area of a triangle (Day 13) · nets & surface area (Day 14) · area of a trapezoid (Day 15), with mixed real-world problems. **Standards:** 6.G.A.1, 6.G.A.3, 6.G.A.4, 6.EE.B.7, 6.NS.B.3.

Directions: Show ALL work by hand in your math notebook (graph paper is best). Label every answer with units. Figures are not drawn to scale. **Watch out:** slanted sides are distractors — only perpendicular heights go in an area formula.

Section A · Area of a Triangle (Day 13)

1. Find the area of the triangle.

Answer: _____



2. A right triangle has legs of 9 in and 12 in. Find its area.

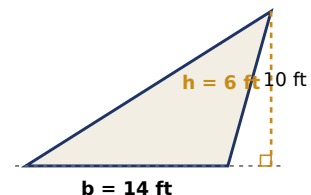
Answer: _____

3. Find the area of a triangle with base 22 m and height 9 m.

Answer: _____

4. **Distractor check.** An obtuse triangle is labeled base 14 ft, height 6 ft, slanted side 10 ft. Find the area.

Answer: _____



5. A triangle has an area of 72 sq cm and a base of 16 cm. Find its height.

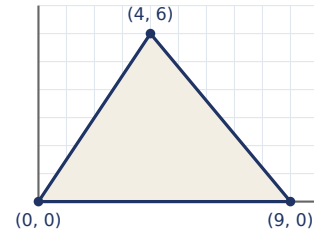
Answer: _____

6. A triangle has an area of 45 sq in and a height of 9 in. Find its base.

Answer: _____

7. Find the area of the triangle plotted on the grid.

Answer: _____



8. **Find the mistake.** For a triangle with base 20 m and height 7 m, Dante wrote $A = 20 \times 7 = 140$ sq m. What did he forget?

Answer: _____

9. A flag is a rectangle 9 in by 6 in with a triangle cut out: base 9 in, height 4 in. How much fabric is left?

Answer: _____

10. A triangular sail has a base of 12 ft and a height of 15 ft. Sailcloth costs \$6 per square foot. What is the cost?

Answer: _____

11. Find the area of a triangle with base 6.4 cm and height 5 cm.

Answer: _____

12. Find the area of a triangle with base $4\frac{1}{2}$ in and height 8 in.

Answer: _____

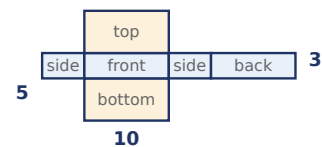
Section B · Nets & Surface Area (Day 14)

13. Find the surface area of a cube with edge 9 cm.

Answer: _____

14. Find the surface area of the rectangular prism shown by this net: 10 in long, 5 in wide, 3 in tall.

Answer: _____



15. A rectangular prism is 7 ft by 4 ft by 6 ft. Find its surface area.

Answer: _____

16. A triangular prism is 12 cm long. Its ends are right triangles with legs 5 cm and 12 cm and hypotenuse 13 cm. Find the surface area.

Answer: _____

17. Find the mistake. For a prism 8 m by 3 m by 5 m, Rosa wrote $SA = (8 \times 3) + (8 \times 5) + (3 \times 5) = 24 + 40 + 15 = 79$ sq m. What went wrong?

Answer: _____

18. A cube has a surface area of 150 sq in. How long is each edge?

Answer: _____

19. A rectangular prism is 6 cm by 4 cm by h cm and has a surface area of 148 sq cm. Find h .

Answer: _____

20. An open-top box (no lid) is 14 in long, 8 in wide, and 5 in tall. How much cardboard does it use?

Answer: _____

21. Surface area is measured in square units, but volume is measured in cubic units. Explain the difference in one sentence, using a gift box.

Answer: _____

22. How many faces, and how many matching pairs, does the net of a rectangular prism have?

Answer: _____

23. A shipping crate is 10 ft by 6 ft by 4 ft. One can of sealant covers 40 sq ft. How many whole cans are needed for the outside?

Answer: _____

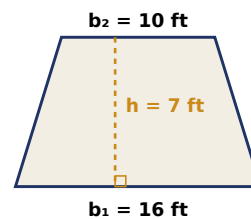
24. Find the surface area of a box that is 3.5 m by 2 m by 4 m.

Answer: _____

Section C · Area of a Trapezoid (Day 15)

25. Find the area of the trapezoid.

Answer: _____

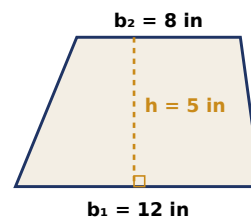


26. A trapezoid has bases 13 cm and 7 cm and a height of 8 cm. Find its area.

Answer: _____

27. **Distractor check.** A trapezoid shows bases 12 in and 8 in, height 5 in, and a slanted side of 6 in. Find the area.

Answer: _____



28. Solve by decomposing: a trapezoid has bases 18 m and 12 m and height 6 m. Give the rectangle area, the triangle area, and the total.

Answer: _____

29. A trapezoid has an area of 96 sq in, bases 9 in and 15 in. Find its height.

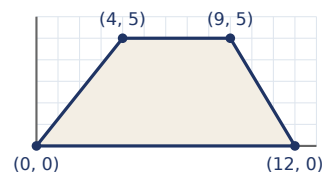
Answer: _____

30. A trapezoid has an area of 110 sq cm, one base of 12 cm, and a height of 10 cm. Find the other base.

Answer: _____

31. Find the area of the trapezoid plotted on the grid.

Answer: _____



32. **Find the mistake.** For a trapezoid with bases 14 ft and 6 ft and height 5 ft, Kevin wrote $(14 + 6) \times 5 = 100$ sq ft. What went wrong?

Answer: _____

33. A window is a trapezoid with bases 5.5 ft and 3.5 ft and height 4 ft. Find its area.

Answer: _____

34. A tabletop is a trapezoid with bases $6\frac{1}{2}$ ft and $3\frac{1}{2}$ ft and height 4 ft. Find its area.

Answer: _____

35. A patio is a rectangle 15 ft by 8 ft plus a trapezoid with bases 15 ft and 9 ft and height 6 ft. Find the total area.

Answer: _____

36. Which has the greater area: a trapezoid with bases 10 m and 6 m and height 8 m, or a triangle with base 16 m and height 9 m? By how much?

Answer: _____

Section D • Mixed Review & Real-World (Days 13-15)

37. A tent's front is a triangle with base 8 ft and height 5 ft. Its floor is a rectangle 8 ft by 10 ft. Find the total of both areas.

Answer: _____

38. A gift box is 12 in by 8 in by 3 in. Wrapping paper comes in sheets of 300 sq in. Is one sheet enough? Show why.

Answer: _____

39. A garden is shaped like a trapezoid with bases 20 ft and 14 ft and height 10 ft. Mulch covers 25 sq ft per bag. How many whole bags are needed?

Answer: _____

40. Name the shape. For each figure, which formula do you use? (a) a shape with four sides and exactly one pair of parallel sides; (b) a shape with three sides; (c) a box's outside; (d) a leaning four-sided shape with two pairs of parallel sides.

Answer: _____

41. A triangle and a trapezoid both have a height of 6 in. The triangle's base is 14 in; the trapezoid's bases are 5 in and 9 in. Show that they have the same area.

Answer: _____

42. Challenge. A storage shed's side wall is a rectangle 10 ft wide and 7 ft tall topped by a triangle 10 ft wide and 4 ft tall. There are two identical side walls. Paint covers 90 sq ft per gallon. How many whole gallons are needed for both walls?

Answer: _____

Answer Key & In-Depth Solutions

Day 16 Review · Days 13-15 · Teacher / family reference — try every problem before checking!

Section A · Area of a Triangle (Day 13)

1. Find the area of the triangle.

- Step 1: $A = \frac{1}{2} \times b \times h = \frac{1}{2} \times 16 \times 7$.
- Step 2: $16 \times 7 = 112$.
- Step 3: $112 \div 2 = 56$.

✓ **Answer: 56 sq cm**

2. A right triangle has legs of 9 in and 12 in. Find its area.

- Step 1: In a right triangle the two legs are the base and the height.
- Step 2: $\frac{1}{2} \times 9 \times 12 = \frac{1}{2} \times 108 = 54$.

✓ **Answer: 54 sq in**

3. Find the area of a triangle with base 22 m and height 9 m.

- Step 1: $\frac{1}{2} \times 22 \times 9$.
- Step 2: $22 \times 9 = 198$.
- Step 3: $198 \div 2 = 99$.

✓ **Answer: 99 sq m**

4. **Distractor check.** An obtuse triangle is labeled base 14 ft, height 6 ft, slanted side 10 ft. Find the area.

- Step 1: Use the base and the perpendicular height only: 14 and 6.
- Step 2: $\frac{1}{2} \times 14 \times 6 = \frac{1}{2} \times 84 = 42$.
- Step 3: The 10 ft slanted side is a trap ($14 \times 10 \div 2 = 70$ is wrong).

✓ **Answer: 42 sq ft**

5. A triangle has an area of 72 sq cm and a base of 16 cm. Find its height.

- Step 1: Double the area first: $144 = 16 \times h$.
- Step 2: $h = 144 \div 16 = 9$.

✓ **Answer: 9 cm**

6. A triangle has an area of 45 sq in and a height of 9 in. Find its base.

- Step 1: Double: $90 = b \times 9$.
- Step 2: $b = 90 \div 9 = 10$.

✓ **Answer: 10 in**

7. Find the area of the triangle plotted on the grid.

- Step 1: Base from (0, 0) to (9, 0): 9 units.
- Step 2: Height: straight up to $y = 6 \rightarrow 6$ units.
- Step 3: $\frac{1}{2} \times 9 \times 6 = 27$.

✓ **Answer: 27 square units**

8. Find the mistake. For a triangle with base 20 m and height 7 m, Dante wrote $A = 20 \times 7 = 140$ sq m. What did he forget?

- Step 1: Dante found the whole parallelogram, not the triangle.
- Step 2: Half of 140 is 70.
- Step 3: He left out the $\frac{1}{2}$.

✓ **Answer: He left out the $\frac{1}{2}$; correct area is 70 sq m**

9. A flag is a rectangle 9 in by 6 in with a triangle cut out: base 9 in, height 4 in. How much fabric is left?

- Step 1: Rectangle: $9 \times 6 = 54$.
- Step 2: Triangle: $\frac{1}{2} \times 9 \times 4 = 18$.
- Step 3: $54 - 18 = 36$.

✓ **Answer: 36 sq in**

10. A triangular sail has a base of 12 ft and a height of 15 ft. Sailcloth costs \$6 per square foot. What is the cost?

- Step 1: Area: $\frac{1}{2} \times 12 \times 15 = \frac{1}{2} \times 180 = 90$ sq ft.
- Step 2: Cost: $90 \times \$6 = \540 .

✓ **Answer: 90 sq ft; \$540**

11. Find the area of a triangle with base 6.4 cm and height 5 cm.

- Step 1: $\frac{1}{2} \times 6.4 \times 5$.
- Step 2: $6.4 \times 5 = 32$.
- Step 3: $32 \div 2 = 16$.

✓ **Answer: 16 sq cm**

12. Find the area of a triangle with base $4\frac{1}{2}$ in and height 8 in.

- Step 1: $\frac{1}{2} \times 4\frac{1}{2} \times 8$.
- Step 2: $4\frac{1}{2} \times 8 = 36$ ($4 \times 8 = 32$ and $\frac{1}{2} \times 8 = 4$).
- Step 3: $36 \div 2 = 18$.

✓ **Answer: 18 sq in**

Section B • Nets & Surface Area (Day 14)

13. Find the surface area of a cube with edge 9 cm.

- Step 1: One face: $9 \times 9 = 81$.
- Step 2: 6 faces: $6 \times 81 = 486$.

✓ **Answer: 486 sq cm**

14. Find the surface area of the rectangular prism shown by this net: 10 in long, 5 in wide, 3 in tall.

- Step 1: $2(10 \times 5) = 100$.
- Step 2: $2(10 \times 3) = 60$.
- Step 3: $2(5 \times 3) = 30$.
- Step 4: $100 + 60 + 30 = 190$.

✓ **Answer: 190 sq in**

15. A rectangular prism is 7 ft by 4 ft by 6 ft. Find its surface area.

- Step 1: $2(7 \times 4) = 56$.
- Step 2: $2(7 \times 6) = 84$.
- Step 3: $2(4 \times 6) = 48$.
- Step 4: $56 + 84 + 48 = 188$.

✓ **Answer: 188 sq ft**

16. A triangular prism is 12 cm long. Its ends are right triangles with legs 5 cm and 12 cm and hypotenuse 13 cm. Find the surface area.

- Step 1: Two triangles: $2 \times (\frac{1}{2} \times 5 \times 12) = 2 \times 30 = 60$.
- Step 2: Triangle perimeter: $5 + 12 + 13 = 30$ cm.
- Step 3: Rectangles: $30 \times 12 = 360$.
- Step 4: $60 + 360 = 420$.

✓ **Answer: 420 sq cm**

17. Find the mistake. For a prism 8 m by 3 m by 5 m, Rosa wrote $SA = (8 \times 3) + (8 \times 5) + (3 \times 5) = 24 + 40 + 15 = 79$ sq m. What went wrong?

- Step 1: Rosa counted only one face from each matching pair.
- Step 2: Every face has a twin: double each product.
- Step 3: $2(24) + 2(40) + 2(15) = 48 + 80 + 30 = 158$.

✓ **Answer: She forgot the matching faces; correct SA is 158 sq m**

18. A cube has a surface area of 150 sq in. How long is each edge?

- Step 1: One face: $150 \div 6 = 25$ sq in.
- Step 2: The face is a square: $5 \times 5 = 25$.
- Step 3: Edge = 5 in.

✓ **Answer: 5 in**

19. A rectangular prism is 6 cm by 4 cm by h cm and has a surface area of 148 sq cm. Find h .

- Step 1: Top and bottom: $2(6 \times 4) = 48$.
- Step 2: Other four faces: $2(6h) + 2(4h) = 12h + 8h = 20h$.
- Step 3: $48 + 20h = 148 \rightarrow 20h = 100 \rightarrow h = 5$.
- Step 4: Check: $48 + 2(30) + 2(20) = 48 + 60 + 40 = 148$. ✓

✓ **Answer: $h = 5$ cm**

20. An open-top box (no lid) is 14 in long, 8 in wide, and 5 in tall. How much cardboard does it use?

- Step 1: Bottom: $14 \times 8 = 112$.
- Step 2: Front and back: $2(14 \times 5) = 140$.
- Step 3: Two ends: $2(8 \times 5) = 80$.
- Step 4: $112 + 140 + 80 = 332$.

✓ **Answer: 332 sq in**

21. Surface area is measured in square units, but volume is measured in cubic units. Explain the difference in one sentence, using a gift box.

- Step 1: Surface area counts the flat outside faces — the wrapping paper — so it is square units.
- Step 2: Volume counts the space inside — how much fits in the box — so it is cubic units.

✓ **Answer: Surface area = paper covering the outside (square units); volume = space inside (cubic units)**

22. How many faces, and how many matching pairs, does the net of a rectangular prism have?

- Step 1: 6 faces.
- Step 2: They form 3 matching pairs: top/bottom, front/back, and the two ends.

✓ **Answer: 6 faces, 3 matching pairs**

23. A shipping crate is 10 ft by 6 ft by 4 ft. One can of sealant covers 40 sq ft. How many whole cans are needed for the outside?

- Step 1: $2(10 \times 6) = 120$.
- Step 2: $2(10 \times 4) = 80$.
- Step 3: $2(6 \times 4) = 48$.
- Step 4: $SA = 120 + 80 + 48 = 248$ sq ft.
- Step 5: $248 \div 40 = 6.2$, so round up.

✓ **Answer: 7 cans (248 sq ft)**

24. Find the surface area of a box that is 3.5 m by 2 m by 4 m.

- Step 1: $2(3.5 \times 2) = 2(7) = 14$.
- Step 2: $2(3.5 \times 4) = 2(14) = 28$.
- Step 3: $2(2 \times 4) = 16$.
- Step 4: $14 + 28 + 16 = 58$.

✓ **Answer: 58 sq m**

Section C · Area of a Trapezoid (Day 15)

25. Find the area of the trapezoid.

- Step 1: Add the bases: $16 + 10 = 26$.
- Step 2: $26 \times 7 = 182$.
- Step 3: $182 \div 2 = 91$.

✓ **Answer: 91 sq ft**

26. A trapezoid has bases 13 cm and 7 cm and a height of 8 cm. Find its area.

- Step 1: $13 + 7 = 20$.
- Step 2: $20 \times 8 = 160$.
- Step 3: $160 \div 2 = 80$.

✓ **Answer: 80 sq cm**

27. Distractor check. A trapezoid shows bases 12 in and 8 in, height 5 in, and a slanted side of 6 in. Find the area.

- Step 1: Only the two parallel bases and the perpendicular height are used.
- Step 2: $\frac{1}{2} \times (12 + 8) \times 5 = \frac{1}{2} \times 100 = 50$.
- Step 3: The 6 in slanted side is a distractor.

✓ **Answer: 50 sq in**

28. Solve by decomposing: a trapezoid has bases 18 m and 12 m and height 6 m. Give the rectangle area, the triangle area, and the total.

- Step 1: Middle rectangle: $12 \times 6 = 72$.
- Step 2: Two end triangles together: base $18 - 12 = 6$, height 6 $\rightarrow \frac{1}{2} \times 6 \times 6 = 18$.
- Step 3: Total: $72 + 18 = 90$.
- Step 4: Formula check: $\frac{1}{2} \times (18 + 12) \times 6 = 90$. ✓

✓ **Answer: 90 sq m**

29. A trapezoid has an area of 96 sq in, bases 9 in and 15 in. Find its height.

- Step 1: Double: $192 = (9 + 15) \times h = 24h$.
- Step 2: $h = 192 \div 24 = 8$.

✓ **Answer: 8 in**

30. A trapezoid has an area of 110 sq cm, one base of 12 cm, and a height of 10 cm. Find the other base.

- Step 1: Double: $220 = (12 + b_2) \times 10$.
- Step 2: $220 \div 10 = 22$, so $12 + b_2 = 22$.
- Step 3: $b_2 = 10$.

✓ **Answer: 10 cm**

31. Find the area of the trapezoid plotted on the grid.

- Step 1: Bottom base: (0, 0) to (12, 0) = 12 units.
- Step 2: Top base: (4, 5) to (9, 5) = 5 units.
- Step 3: Height = 5 units.
- Step 4: $\frac{1}{2} \times (12 + 5) \times 5 = \frac{1}{2} \times 85 = 42.5$.

✓ **Answer: 42.5 square units**

32. Find the mistake. For a trapezoid with bases 14 ft and 6 ft and height 5 ft, Kevin wrote $(14 + 6) \times 5 = 100$ sq ft. What went wrong?

- Step 1: Kevin found the doubled parallelogram.
- Step 2: One trapezoid is half: $100 \div 2 = 50$.
- Step 3: He forgot to cut it in half.

✓ **Answer: He left out the $\frac{1}{2}$; correct area is 50 sq ft**

33. A window is a trapezoid with bases 5.5 ft and 3.5 ft and height 4 ft. Find its area.

- Step 1: $5.5 + 3.5 = 9$.
- Step 2: $9 \times 4 = 36$.
- Step 3: $36 \div 2 = 18$.

✓ **Answer: 18 sq ft**

34. A tabletop is a trapezoid with bases $6\frac{1}{2}$ ft and $3\frac{1}{2}$ ft and height 4 ft. Find its area.

- Step 1: $6\frac{1}{2} + 3\frac{1}{2} = 10$.
- Step 2: $10 \times 4 = 40$.
- Step 3: $40 \div 2 = 20$.

✓ **Answer: 20 sq ft**

35. A patio is a rectangle 15 ft by 8 ft plus a trapezoid with bases 15 ft and 9 ft and height 6 ft. Find the total area.

- Step 1: Rectangle: $15 \times 8 = 120$.
- Step 2: Trapezoid: $\frac{1}{2} \times (15 + 9) \times 6 = \frac{1}{2} \times 144 = 72$.
- Step 3: $120 + 72 = 192$.

✓ **Answer: 192 sq ft**

36. Which has the greater area: a trapezoid with bases 10 m and 6 m and height 8 m, or a triangle with base 16 m and height 9 m? By how much?

- Step 1: Trapezoid: $\frac{1}{2} \times (10 + 6) \times 8 = \frac{1}{2} \times 128 = 64$.
- Step 2: Triangle: $\frac{1}{2} \times 16 \times 9 = \frac{1}{2} \times 144 = 72$.
- Step 3: $72 - 64 = 8$.

✓ **Answer: The triangle, by 8 sq m**

Section D • Mixed Review & Real-World (Days 13–15)

37. A tent's front is a triangle with base 8 ft and height 5 ft. Its floor is a rectangle 8 ft by 10 ft. Find the total of both areas.

- Step 1: Triangle: $\frac{1}{2} \times 8 \times 5 = 20$.
- Step 2: Rectangle: $8 \times 10 = 80$.
- Step 3: $20 + 80 = 100$.

✓ **Answer: 100 sq ft**

38. A gift box is 12 in by 8 in by 3 in. Wrapping paper comes in sheets of 300 sq in. Is one sheet enough? Show why.

- Step 1: $2(12 \times 8) = 192$.
- Step 2: $2(12 \times 3) = 72$.
- Step 3: $2(8 \times 3) = 48$.
- Step 4: $SA = 192 + 72 + 48 = 312$ sq in.
- Step 5: $312 > 300$, so one sheet is not enough.

✓ **Answer: No — it needs 312 sq in, and a sheet is only 300 sq in**

39. A garden is shaped like a trapezoid with bases 20 ft and 14 ft and height 10 ft. Mulch covers 25 sq ft per bag. How many whole bags are needed?

- Step 1: Area: $\frac{1}{2} \times (20 + 14) \times 10 = \frac{1}{2} \times 340 = 170$ sq ft.
- Step 2: $170 \div 25 = 6.8$.
- Step 3: Round up to 7.

✓ **Answer: 7 bags (170 sq ft)**

40. Name the shape. For each figure, which formula do you use? (a) a shape with four sides and exactly one pair of parallel sides; (b) a shape with three sides; (c) a box's outside; (d) a leaning four-sided shape with two pairs of parallel sides.

- Step 1: (a) Trapezoid $\rightarrow A = \frac{1}{2}(b_1 + b_2)h$.
- Step 2: (b) Triangle $\rightarrow A = \frac{1}{2}bh$.
- Step 3: (c) Surface area $\rightarrow$ add the area of every face.
- Step 4: (d) Parallelogram $\rightarrow A = bh$.

✓ **Answer: (a) $\frac{1}{2}(b_1+b_2)h$ • (b) $\frac{1}{2}bh$ • (c) sum of all faces • (d) $b \times h$**

41. A triangle and a trapezoid both have a height of 6 in. The triangle's base is 14 in; the trapezoid's bases are 5 in and 9 in. Show that they have the same area.

- Step 1: Triangle: $\frac{1}{2} \times 14 \times 6 = 42$.
- Step 2: Trapezoid: $\frac{1}{2} \times (5 + 9) \times 6 = \frac{1}{2} \times 84 = 42$.
- Step 3: Both are 42 sq in — because $5 + 9 = 14$, the two shapes use the same numbers in the same formula.

✓ **Answer: Both are 42 sq in**

42. Challenge. A storage shed's side wall is a rectangle 10 ft wide and 7 ft tall topped by a triangle 10 ft wide and 4 ft tall. There are two identical side walls. Paint covers 90 sq ft per gallon. How many whole gallons are needed for both walls?

- Step 1: One wall: rectangle $10 \times 7 = 70$; triangle $\frac{1}{2} \times 10 \times 4 = 20$; total 90 sq ft.
- Step 2: Two walls: $2 \times 90 = 180$ sq ft.
- Step 3: $180 \div 90 = 2$ exactly.

✓ **Answer: 2 gallons (180 sq ft)**